ML Assignment – 1

Question-1

Question - 1

Given $x_1 = (1, 2, 0)^T \quad y_1 = +1$

$x_2 = (1, 2, 2)^T \quad y_2 = +1$

$x_3 = (1, 0, 2)^T \quad y_3 = -1$

$w_0 = (0, 6, 6)^T$

$\text{Sign}(w^T x_i) = y_i$

1. $\text{Sign}([0, 6, 6]^T [1, 2, 0]) = +1$

$\text{sign}([0 + 12 + 0]) = +1$

$\text{sign}(12) = +1$

So, $\text{sign}(w^T x_1) = y_1$ ✓ satisfied

2. $\text{Sign}([0, 6, 6]^T [1, 2, 2]) = +1$

$\text{Sign}(0, 12, 12) = +1$

$\text{Sign}(24) = +1$

So, $\text{sign}(w^T x_2) = y_2$ ✓ satisfied

3. $\text{Sign}([0, 6, 6)^T [1, 0, 2] = -1$

$\text{sign}(0, 0, 12) = -1$

$\text{Sign}(12) = -1$

$\text{sign}(w^T x_3) = y_3$ ✗ Not satisfied

O weight vector : $w = w + y_3 x_3$

$= (0, 6, 6) + ((-1)(1, 0, 2))$

$= (0, 6, 6) + (-1, 0, -2)$

$= (0 + (-1) + (6 + 0) + (6 + (-2)))$

✓ $w = (-1, 6, 4)$

$\text{sign}(w^T x_3) = (-1, 6, 4)(1, 0, 2)$

$= (-1 + 0 + 8)$

$\text{sign}(+7) = (+7)$

$\text{sig}(+7) = -1$ ✗ Not satisfied

O weight vector: $w = (-1, 6, 4) + (-1)(1, 0, 2)$

$= (-1, 6, 4) + (-1 + 0 + (-2)$

$= ((-1) + (-1) + (6 + 0) + (4 + (-2)$

✓ $w = (-2, 6, 2)$ ~~$w_1 = (-2 + 6 + 2)$~~

~~$\text{sign}(8) = (-1)$ ✗ Not satisfied~~

$\text{sign}(w^T x_3) = (-2, 6, 2)(1, 0, 2)$

$= (-2 + 0 + 4)$

$\text{sign}(2) = -1$ ✗ Not satisfied

∘ weight vector = $w + y_3 x_3$

$(-2, 6, 2) + (-1(1, 0, 2)$

$(-2, 6, 2) + (-1, 0, -2)$

$(-2 + (-1) + (6+0) + (2 + (-2)$

$(-3, 6, 0)$

~~sign(-3) = -1 ✗ not satisfied~~

$\text{sign}(w^T x_3) = (-3, 6, 0)(1, 0, 2)$

$(-3, 0, 0)$

$\text{sign}(-3) = -1$ ✓ satisfied

So,

$\text{sign}(w^T x_1) = y_1 : \text{sign}(12) = +1$

$\text{sign}(w^T x_2) = y_2 = \text{sign}(24) = +1$

$\text{sign}(w^T x_3) = y_3 = \text{sign}(-3) = -1$ ✓✓